

NLP Advance: Hate Speech Detection for Maintaining DEI Businesses, Schools, and Community Outreach

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Specialization: NLP

Problem Description: The problem is to develop and deploy an NLP model that can detect hate speech in text.

Business Application: This model will be used to support pro-DEI businesses and schools that are fighting to protect LGBTQ, Transgender, minority rights.

Data understanding

The dataset provided for the project is a dataset from Kaggle called: [Twitter hate speech](#)

Number of files: 2

Types of files: test data and train data

Names of files: test_tweets_anufFYb8.csv and train_e6o_V3IV.csv

Data Type: Structured .csv files

Training Data Number of Features: 3

Training Data Features: id, label, tweet

Binary label: 0 = no hate, 1=hate

Number of Records: 31,962

Test Data Number of Features: 2

Test Data Features: id, tweet

Number of Records: 17,197

Typically, datasets are split 80:20 training data: test data

The ratio here is much higher: 31,962: 17,197 ~ 2:1 training data: test data

Main function of the model is to label the test data 0 for no hate and 1 for hate

Initial Data Problems in the Data Set :

1. The training data needs to be reviewed record by record to verify the labels are correct
2. The data needs to be converted to plain English text
3. There are symbols like wingdings that are not English text that will need to be removed. They look like they may be redactions of names
4. There are '#' and '@user' symbols that will also need to be removed.
5. There are abbreviations that will have to be handled such as otw = on the way or btw = by the way, etc.
6. The tweets may not be representative of good contrasting hate and no hate for training purposes
7. The files may need to be augmented with more extreme versions of hate and no hate texts for training
8. '@user' may indicate a greater likelihood of hate since it is direct. Will have to evaluate
9. Sarcasm will be difficult to determine.

Missing Data:

1. test_tweets_df : No missing data
2. train_tweets_df: No missing data

Proportion of hate to no hate labels for the training set

Hate (1): ~7%

No Hate (0): ~93%

The data is very skewed toward no hate vs. hate in the training set

Other Problems Expected in Text Data:

1. Outliers in Text Data
2. Skewed Data
3. Noise
4. High dimensional text embeddings

Possible Approaches for Data Problems in Text Data

Outliers in Text Data

1. Token Frequency Analysis: analyze the characters individually and get percentages for each character. Identify rarely used and overused characters and decide whether to normalize them
2. Clustering: Use k-means to detect and isolate outliers outside of clusters
3. Manual Inspection: Evaluate the records flagged by automated systems for small data sets

Skewed Data

1. Augmentation: generate synthetic examples for under-represented classes. In this data problem, there are only 2 classes hate and no hate and hate is under-represented at 7% of the total tweets in the training data set and no hate is over-represented at 93% of the total tweets in the training data set.
2. Resampling with SMOTE
3. Normalization: use transformations or scaling to reduce skewness in numerical features extracted from text like word counts or character counts in the tweets

Text Related Issues

1. Remove Noise: cleanup HTML tags, special characters, nonsense characters and irrelevant tokens.
2. Reduce dimensions of characters: Use PCA or t-SNE
3. Transfer Learning and Feature Engineering: Use feature engineering already figured out by other models to be useful in dealing with text

Approaches That Will Most Likely Be Used

1. Token Frequency Analysis. There may be some way to see that hate text is generated by the @user frequency. Usually people that are generating hate text are directing it toward someone. The same goes true for excessive hashtags.
2. Clustering may need to be used.
3. Manual Inspection will be used to evaluate the data
4. Noise will be removed.
5. Data will likely be cleaned up after the token frequency analysis
6. Augmentation by generating synthetic samples will likely be employed to increase the Hate labeled samples in the training data set
7. SMOTE may be employed

8. Transfer Learning to find other features previous models used will likely be employed

Github Repo link : [Upload files · wunderdux/VC](#)